

TASK 01 – Writing Better Prompts

Prompt Engineering Internship

In this task, three vague or ineffective prompts are rewritten into improved versions with greater clarity, specificity, and appropriate tone. Each improved prompt is designed to guide the AI toward a more useful and relevant response. The outputs are compared to understand how prompt design affects LLM behaviour.

Topic 1: Artificial Intelligence

Type	Content
Vague Prompt	Explain artificial intelligence.
Output	Artificial Intelligence (AI) is the ability of computers and machines to perform tasks that normally require human intelligence. It includes learning, reasoning, problem-solving, understanding language, and recognizing patterns. AI is used in areas such as healthcare, education, transportation, banking, and entertainment.
Improved Prompt	Explain Artificial Intelligence in 5 simple bullet points for a college student who is new to AI. Include its definition, two real-world applications, one benefit, and one limitation. Use simple and easy-to-understand language.
Output	1. Meaning: Artificial Intelligence is a technology that enables computers to perform tasks that usually require human intelligence. 2. Learning: AI systems can learn patterns from data and use them to make predictions or decisions. 3. Applications: AI is used in chatbots, medical diagnosis, recommendation systems, self-driving technologies, and fraud detection. 4. Benefit: AI can automate repetitive tasks and process large amounts of information quickly. 5. Limitation: AI depends heavily on the quality of its training data and

	may produce incorrect or biased results.
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Analysis: The vague prompt produces a general explanation without a defined audience or structure. The improved prompt clearly specifies the audience, number of points, required information, and language style. As a result, the response becomes more focused, organized, and easier for a beginner to understand.

Topic 2: Python Programming

Type	Content
Vague Prompt	Write a Python program.
Output	Python can be used to create many different types of programs. For example, a simple program can take two numbers as input and calculate their sum.
Improved Prompt	Write a beginner-friendly Python program that accepts two numbers from the user and displays their sum, difference, product, and quotient. Use simple Python syntax and add short comments explaining each operation.
Output	<pre>PROGRAM: # Get two numbers from the user num1 = float(input("Enter the first number: ")) num2 = float(input("Enter the second number: ")) # Calculate and display the sum print("Sum:", num1 + num2) # Calculate and display the difference print("Difference:", num1 - num2) # Calculate and display the product print("Product:", num1 * num2) # Calculate and display the quotient</pre>

	<pre> if num2 != 0: print("Quotient:", num1 / num2) else: print("Quotient: Cannot divide by zero") EXAMPLE OUTPUT: Enter the first number: 10 Enter the second number: 5 Sum: 15.0 Difference: 5.0 Product: 50.0 Quotient: 2.0 </pre>
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Analysis: The vague prompt does not explain what type of Python program is required, so the AI has to make assumptions. The improved prompt specifies the input, required operations, programming level, syntax style, and an important condition for division by zero. Therefore, the resulting code is more relevant, complete, and practical.

Topic 3: Internship Preparation

Type	Content
Vague Prompt	Give me interview tips.
Output	To perform well in an interview, prepare for common questions, research the company, dress appropriately, communicate clearly, and be confident. You should also prepare questions to ask the interviewer.
Improved Prompt	Give me 6 practical interview preparation tips for a college student attending a technology internship interview. Include technical preparation, project explanation, communication, confidence, company

	research, and questions to ask the interviewer. Keep each tip within 2 sentences and use a professional but encouraging tone.
Output	1. Prepare Technical Basics: Revise the programming languages, tools, and technical concepts mentioned in your resume. Focus on understanding rather than memorizing answers. 2. Know Your Projects: Be ready to explain your projects, your role, technologies used, and challenges you solved. 3. Practice Communication: Answer questions clearly and avoid unnecessary information. Practice introducing yourself in a short and confident manner. 4. Build Confidence: Maintain good eye contact, listen carefully, and take a moment to think before answering difficult questions. 5. Research the Company: Learn about the company's products, services, culture, and the role you applied for. This shows genuine interest. 6. Prepare Questions: Ask meaningful questions about the internship responsibilities, learning opportunities, and team structure.

Analysis: The vague prompt gives only general advice because it does not define the interview type, audience, or expected format. The improved prompt provides a specific audience, technology-related context, required topics, sentence limit, and tone. This makes the response more practical and directly useful for an internship candidate.

Key Learnings from Task 01

- **Specificity Matters** – A specific prompt gives the AI clear instructions and reduces unnecessary information.
- **Define the Audience** – Mentioning the target audience helps the AI adjust the complexity and vocabulary of the response.
- **Set the Format** – Specifying bullet points, paragraphs, tables, or code makes the output more organized.
- **Add Constraints and Scope** – Requirements such as number of points, sentence limits, and required topics make the response more focused.
- **Tone and Context** – Specifying a professional, friendly, simple, or encouraging tone helps generate a response suitable for the intended purpose.

Conclusion

This task demonstrates that the quality of an AI response depends greatly on how the prompt is written. Vague prompts often produce broad and generic answers, while clear and specific prompts guide the AI toward relevant, structured, and useful results. Effective prompt engineering therefore requires clarity, specificity, audience awareness, context, and appropriate constraints.